

# Reading the evidence

|      |       |      |
|------|-------|------|
| Name | Group | Date |
|------|-------|------|

**Safety** Rusted nails are sharp and rust carries a tetanus risk. TONGS OR GLOVES ONLY — never bare hands. Count your nails out and count them back in. Tell the teacher straight away if anyone gets scratched.

## 1 • Drying — and why we are being this fussy

One drop of water weighs 0.05 g. The mass your nails have actually gained might only be 0.06 g. So a single drop left behind could DOUBLE your result and you would never know. Water is not rust, and the scale cannot tell the difference.

1. Lift your nails out with tongs, keeping each jar’s three together and labelled.
2. Rinse briefly under the tap.
3. Blot hard with paper towel. Then blot again.
4. Spread them on your tray and leave them until they look AND feel completely dry.

## 2 • The numbers

Weigh all three nails together, exactly as you did in Week 1. Record to TWO DECIMAL PLACES, always — 13.80 g, not 13.8 g.

|                    | Jar 1 | Jar 2 | Jar 3 |
|--------------------|-------|-------|-------|
| Mass in Week 1 (g) |       |       |       |
| Mass today (g)     |       |       |       |
| DIFFERENCE (g)     |       |       |       |

**The check that tells you whether to trust any of it** The dry sealed jar should weigh what it weighed in Week 1, give or take a division. If it has GAINED, the drying has failed and every other number in the class is too high.

Did your nails get heavier, lighter, or stay the same?

Is that what you expected five weeks ago? Say what you thought would happen.

Nothing was added to your jar. So where did the extra mass come from?

### 3 • Graph your three jars

One bar per jar. Write your own scale up the left-hand side — look at your biggest number first and choose a scale that fits it.

Mass gained (g)

|                  |       |       |       |
|------------------|-------|-------|-------|
| Write your scale |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  |       |       |       |
|                  | Jar 1 | Jar 2 | Jar 3 |

Are your three bars the same height? Should they be — you set all three up identically?

How different would two jars have to be before you would call it a REAL difference rather than just wobble?

**Why we ran three** If you had only run one jar, you would have no idea whether your number was typical or a fluke. Three tells you how much your own results wobble — and that tells you how big a difference has to be before it means anything.

Mass gained (g)

5. Which condition rusted the most?

6. Which TWO conditions barely rusted at all? What do they have in common — and what is different between them?

7. Compare the half-submerged jar with the fully submerged one. Which won, and were you expecting that?

8. Write the pattern in one sentence. What does rusting need?

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**Push me** Express each gain as a percentage of the starting mass. Does that change which condition won? Then: is the gap BETWEEN conditions bigger than the gap between your own three jars — and what does your answer mean for how much you trust the result?

# Graphing the rust results

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## My three jars

Colour in one bar for each jar. The scale is already drawn for you.

|        |       |       |       |
|--------|-------|-------|-------|
| 0.30 g |       |       |       |
| 0.25 g |       |       |       |
| 0.20 g |       |       |       |
| 0.15 g |       |       |       |
| 0.10 g |       |       |       |
| 0.05 g |       |       |       |
| 0 g    |       |       |       |
|        | Jar 1 | Jar 2 | Jar 3 |

## The whole class

|        |   |   |   |   |   |   |
|--------|---|---|---|---|---|---|
| 0.30 g |   |   |   |   |   |   |
| 0.25 g |   |   |   |   |   |   |
| 0.20 g |   |   |   |   |   |   |
| 0.15 g |   |   |   |   |   |   |
| 0.10 g |   |   |   |   |   |   |
| 0.05 g |   |   |   |   |   |   |
| 0 g    |   |   |   |   |   |   |
|        | A | B | C | D | E | F |

## Finish these sentences

The condition that rusted the MOST was \_\_\_\_\_ .

The conditions that barely rusted were \_\_\_\_\_ .

One of those had no \_\_\_\_\_ and the other had no \_\_\_\_\_ .

This tells me that rusting needs \_\_\_\_\_ and \_\_\_\_\_ at the same time.

My nails got \_\_\_\_\_ because \_\_\_\_\_ .

\_\_\_\_\_ .

**Word bank** heavier · lighter · water · oxygen · air · joined the nail · half-submerged · dry sealed · under oil · new substance

# The swap — somebody else’s data

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**You were not there** This is the other class’s data. You do not know how they did it. They tested something you did not, and you tested something they did not — neither class can answer the client’s question alone. But before you use their numbers, you have to decide whether you trust them.

**WRITE THESE DOWN BEFORE YOU DISCUSS ANYTHING.** Once someone says what they think out loud, it gets very hard to notice anything different.

## 1 • Two things you CANNOT tell from this data

*What did they not write down that you would need to know?*

## 2 • One thing you think may have gone wrong

*Be specific. Name the step.*

## 3 • One question you would ask them

**Things worth looking for** Did they dry the nails the same way? Did every jar start on the same day? Were the jars in the same place — same light, same temperature? Same size nails? Is one of their three jars wildly different from the other two, and did they say why? Did they record to two decimal places or round? Is anything missing?

## Now turn it around

What would the OTHER CLASS say about OUR method? Name two things they could fairly attack.

|   |  |
|---|--|
| 1 |  |
| 2 |  |

### Why this matters

9. Could either class have answered the client's question on its own? Explain.
10. Real scientists pull each other's methods apart constantly. Why would that make science BETTER rather than worse?

**The point** Scientists publish their methods precisely so other people can attack them. A finding that survives other people trying to pull it apart is worth far more than one nobody checked. That is not scientists being difficult with each other — it is the part that makes the answer trustworthy.

**Coming up next** You are about to write your own investigation plan, and another group is about to do exactly this to it. What would YOU attack?